

★ WEEK 1 · PART 1 of 3

From Prices to Returns

The Language of Financial Markets

Simple vs log · arithmetic vs geometric · when to use which · what can go wrong

Before any model, we must speak the market's native unit: the return. This part builds that vocabulary carefully — every formula, its assumptions, and its traps.

1

PART 1 • RETURNS

Returns: The Building Block of Everything

Prices are what we observe. Returns are what we model. This part is the foundation the whole course stands on.

Why We Model Returns, Not Prices

Three reasons a professional never models the raw price series

1

Prices are non-stationary — returns are (nearly) stationary

Price levels trend and drift with no fixed mean or variance. Most statistical tools assume a stable distribution; returns provide it.

2

Returns are comparable across assets and scale

A \$10 move means one thing on a \$50 stock and another on a \$5,000 one. A 2% return means the same thing everywhere.

3

Returns compound cleanly — prices do not

Growth of wealth is a product of gross returns. Working in returns lets us aggregate over time and across a portfolio with simple algebra.

Rule of thumb: observe prices, but always convert to returns before you analyze.

What Is a Financial Time Series?

A single realized path drawn from an underlying random process

A time series is a sequence of random variables ordered in time. The data we see is ONE realized path — not the process itself.

The mental shift

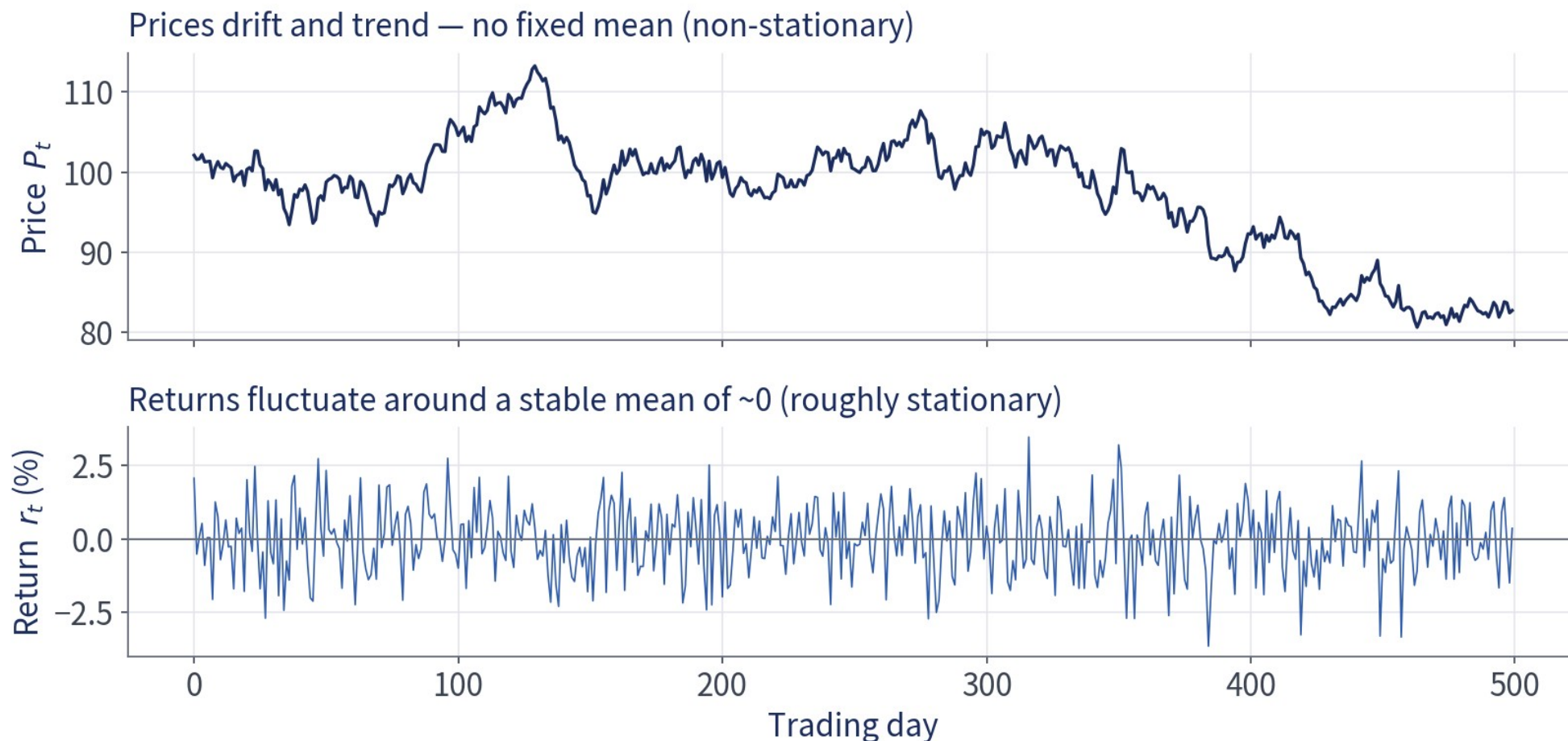
- We never see the true data-generating process.
- We see one sample path and must infer the rules behind it.
- Order matters: yesterday can inform today.

Why this is hard

- Only one history exists — no repeated experiments.
- The rules can change over time (regimes).
- This is why finance leans on stylized facts, not certainty.

Prices Trend; Returns Fluctuate Around Zero

Top: the price wanders with no fixed level. Bottom: its returns hover around a stable mean — this is why we model returns.



Simple (Arithmetic) Return

The intuitive percentage change over one period

$$R_t = \frac{P_t - P_{t-1}}{P_{t-1}} = \frac{P_t}{P_{t-1}} - 1$$

What it answers

- By what percent did my money grow this period?
- Directly interpretable: +3% means +3 cents on the dollar.
- Adds correctly ACROSS assets in a portfolio.

Where it bites

- Does NOT add across time — you must multiply.
- Bounded below by -100% (you cannot lose more than all).
- Right-skewed, so its distribution is awkward to model.

Gross Return and the +1 Convention

The quantity that actually compounds

$$1 + R_t = \frac{P_t}{P_{t-1}} \quad (\text{gross return})$$

Why we add the 1

- Gross return = the growth multiplier of one dollar.
- A +5% return becomes a gross return of 1.05.
- Wealth after k periods = product of gross returns.

A useful habit

- Think in multipliers, not percentages, when compounding.
- $1.05 \times 0.95 = 0.9975 \rightarrow$ a net loss, not break-even.
- This preview is the +20% / -20% trap we meet shortly.

Compounding Simple Returns Is Multiplicative

Over many periods, gross returns multiply — they never simply add

$$1 + R_t(k) = \prod_{j=0}^{k-1} (1 + R_{t-j}) \quad (\text{multiplicative})$$

The correct rule

- Multi-period gross return = product of each period's gross return.
- You cannot sum simple returns to get a multi-period total.

Common mistake

- Adding daily simple returns to get a monthly figure.
- The error grows with volatility and horizon.
- Log returns will fix exactly this problem.

Log (Continuously Compounded) Return

The modeler's default unit

$$r_t = \ln \frac{P_t}{P_{t-1}} = \ln(1 + R_t)$$

Why modelers love it

- Adds cleanly over time (next slide).
- Symmetric and unbounded — friendlier for the normal model.
- The natural unit for continuous-time finance.

The price you pay

- Slightly less intuitive than a plain percentage.
- Does NOT add across assets in a portfolio.
- So convert back to simple for portfolio arithmetic.

Log Returns Add Through Time

The single most convenient property in time-series finance

$$r_t(k) = \sum_{j=0}^{k-1} r_{t-j} \quad (\text{additive through time})$$

Why it works

- $\log(P_T / P_0)$ telescopes into a sum of one-period logs.
- A k -day log return is just the sum of k daily log returns.

What it buys us

- Time-aggregation becomes simple addition.
- Means and variances scale in a clean, tractable way.
- This underpins the annualization rule later in this part.

Simple and Log Returns Are Two Views of One Move

An exact, invertible relationship — not an approximation

$$r_t = \ln(1 + R_t) \quad \Longleftrightarrow \quad R_t = e^{r_t} - 1$$

How to read it

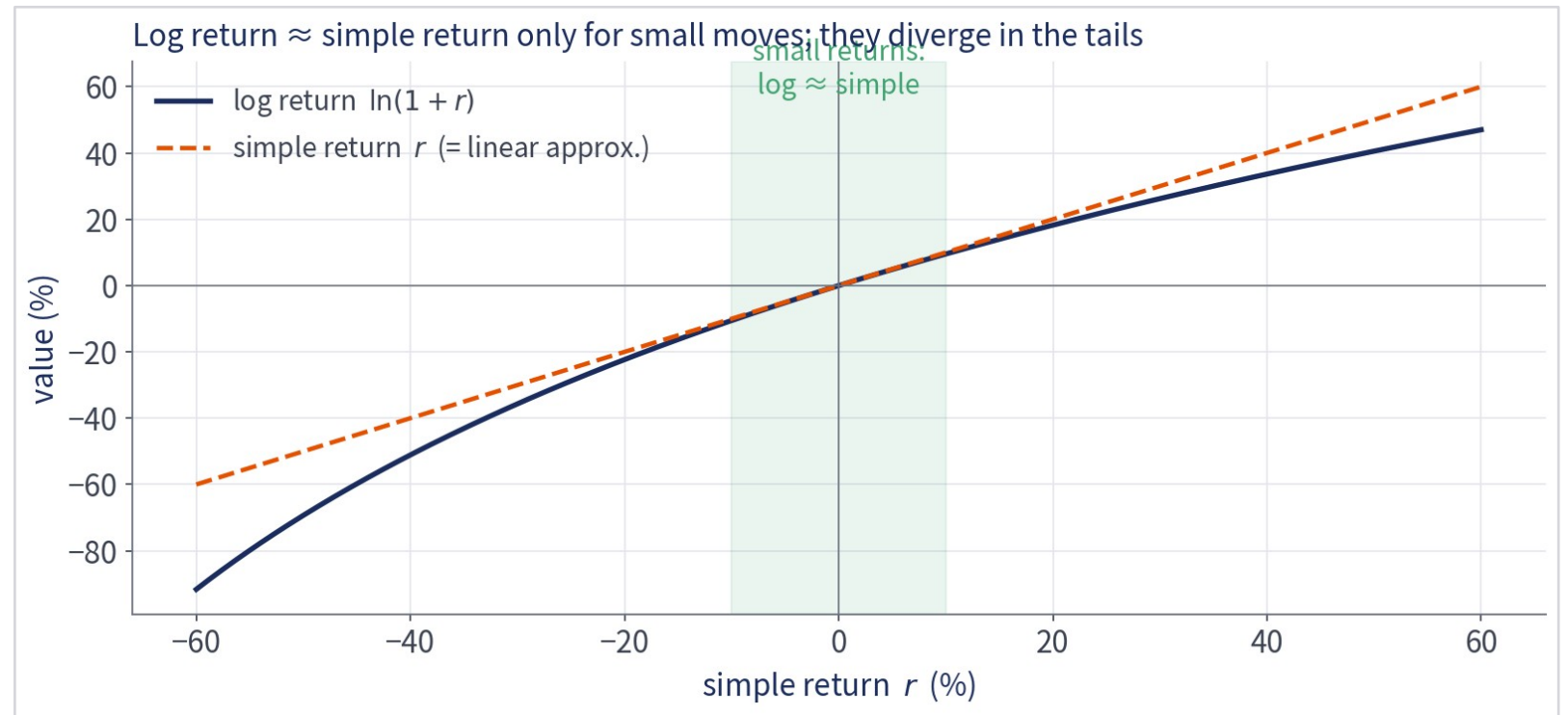
- Given either return, you can recover the other exactly — no information is lost in the conversion.
- Log return is always slightly smaller than the simple return (except at zero), because $\ln(1+R) < R$ for $R > 0$.
- The gap between them is negligible for small moves and grows for large ones — which the next chart makes visual.

When Is $\text{Log} \approx \text{Simple}$?

The approximation is excellent for small returns and breaks down in the tails.

Reading the chart

- Green band: $|r| < 10\%$.
- There the two curves overlap.
- Daily equity moves live here.
- Tails: log bends below simple.
- So for crashes, the choice matters.



Computing Both Returns by Hand

A two-day price path — follow every digit

Day	Price P_t	Simple $R_t = \frac{P_t}{P_{t-1}} - 1$	Log $r_t = \ln(P_t/P_{t-1})$
0	100.00	—	—
1	110.00	+10.00%	+9.531%
2	99.00	−10.00%	−10.536%
0→2 total	99.00	−1.00% ($=1.10 \times 0.90 - 1$)	−1.005% ($=+9.531 - 10.536$)

The last row is the point: simple returns multiply ($\rightarrow -1\%$), while log returns add ($\rightarrow -1.005\%$).

Averaging Returns, Way 1: Arithmetic Mean

The simple average of period returns

$$\bar{R}_A = \frac{1}{T} \sum_{t=1}^T R_t \quad (\text{arithmetic mean})$$

What it estimates

- The expected return of a SINGLE future period.
- The right input for forward-looking, one-period models.
- Used in CAPM, mean-variance optimization, forecasts.

The catch

- It overstates the growth you actually realize over time.
- Because it ignores compounding of gains and losses.
- For realized growth, use the geometric mean →

Averaging Returns, Way 2: Geometric Mean

The constant rate that reproduces the actual end wealth

$$R_G = \left[\prod_{t=1}^T (1 + R_t) \right]^{1/T} - 1 \quad (\text{geometric mean})$$

What it estimates

- The realized compound growth rate over the whole path.
- The honest number for 'how did this investment do?'
- Equivalent to the average of log returns.

When to report it

- Historical performance and back-tests.
- Comparing funds over the same past window.
- Never larger than the arithmetic mean (next slide).

Arithmetic $\geq$ Geometric: The Volatility Drag

The two means differ by roughly half the variance

$$R_G \approx \bar{R}_A - \frac{1}{2}\sigma^2 \leq \bar{R}_A$$

The intuition

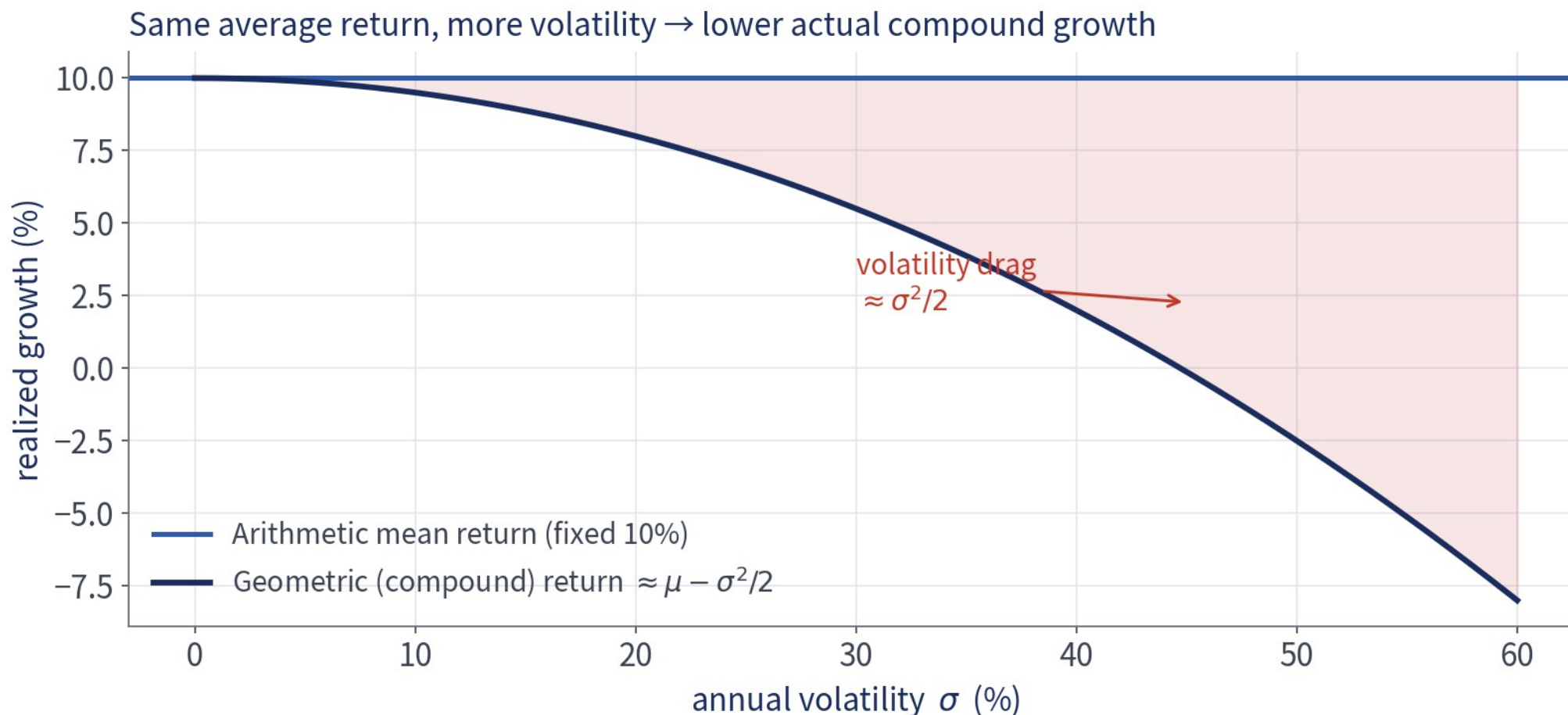
- A -50% loss needs a +100% gain to recover.
- Volatility hurts compound growth asymmetrically.
- More volatility $\rightarrow$ bigger gap between the two means.

The practical warning

- Quoting the arithmetic mean as 'your return' overstates it.
- The drag is about $\sigma^2 / 2$ per period.
- The next chart shows the gap widening with volatility.

Same Average, Less Growth: Volatility Drag

Hold the arithmetic mean fixed at 10%. As volatility rises, the realized compound return falls away by about $\sigma^2/2$.



The +20% / -20% Trap

Why equal-looking gains and losses do not cancel

$$(1 + 0.20)(1 - 0.20) - 1 = 0.96 - 1 = -4\%$$

What just happened

- Up 20% then down 20% leaves you DOWN 4%, not flat.
- Simple returns cannot be added — they compound.
- The loss acts on a larger base than the gain did.

The log-return view

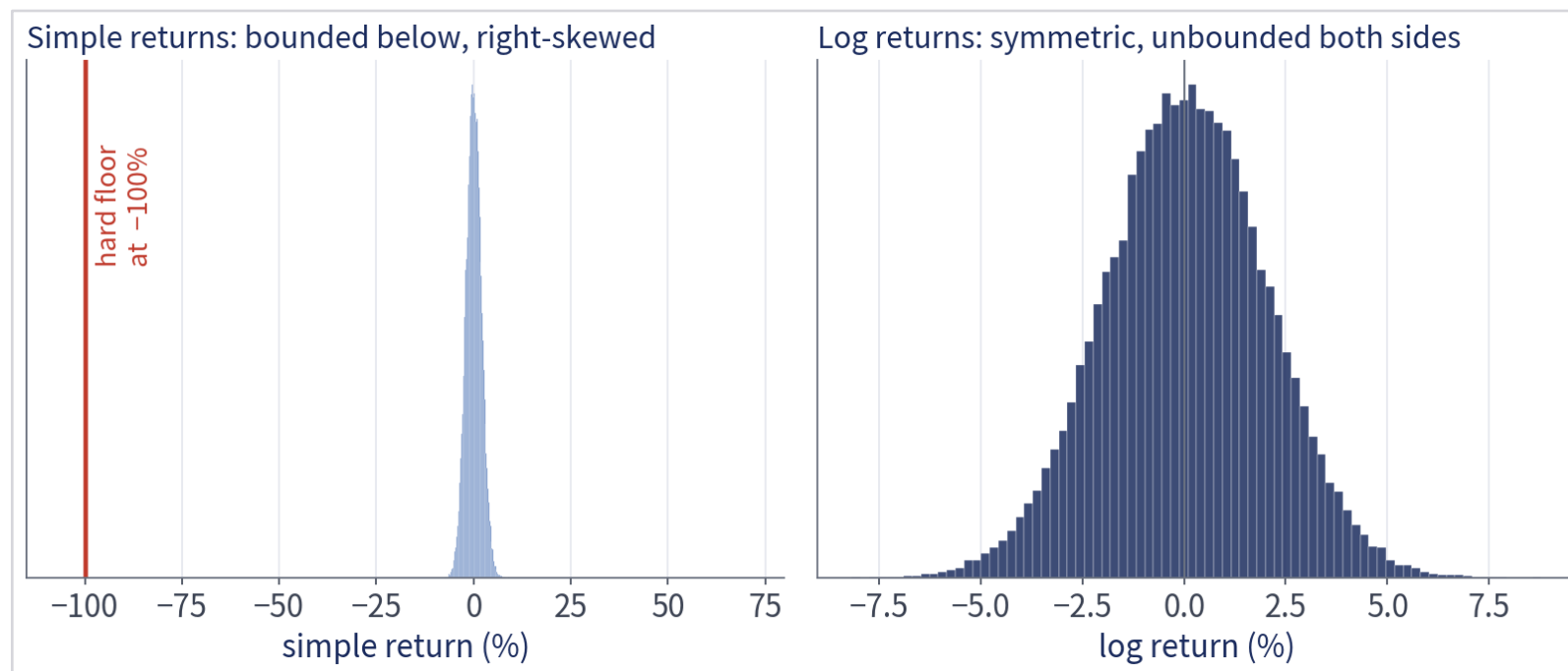
- In logs: +18.23% then -22.31% = -4.08% $\approx \ln(0.96)$.
- Logs add, so the arithmetic is transparent.
- This is exactly why we prefer logs over time.

Why Log Is the Modeling Default

The distributional shapes differ — and one is far friendlier to standard statistics.

What to notice

- Simple: floor at -100% .
- So it is right-skewed.
- Log: symmetric, no floor.
- Closer to a normal shape.
- Hence the modeling default.



Annualizing: The Square-Root-of-Time Rule

Under i.i.d. returns, the mean scales with time but volatility scales with its square root.

The rule

- Mean: multiply by h .
- Vol: multiply by $\sqrt{h}$.
- 1 year ≈ 252 days.
- So $\sigma_{yr} \approx 16 \times \sigma_{day}$.
- Assumes independence.



Real, Nominal, and Excess Returns

Three adjustments you must make before comparing returns fairly

$$1 + R_t^{\text{real}} = \frac{1 + R_t^{\text{nom}}}{1 + \pi_t}, \quad R_t^e = R_t - R_t^f$$

Real vs nominal

- Nominal returns ignore inflation.
- Real return strips out the change in the price level.
- Compare purchasing power, not headline numbers.

Excess return

- Subtract the risk-free rate to isolate the risk premium.
- This is what asset-pricing models actually explain.
- Also remember total return: add reinvested dividends.

Which Return Should You Use?

Across assets → simple. Across time → log. For an average: arithmetic looks forward, geometric looks back.

Which return should you use?

The answer depends on whether you are adding across assets or across time.

Adding ACROSS assets (a portfolio at one date)

Use SIMPLE returns

Portfolio return = Sum of (weight x simple return).
Log returns are NOT additive across assets.

Adding OVER time (many periods compounding)

Use LOG returns

T-period log return = Sum of one-period log returns.
They add up cleanly through time.

Summarizing "the average return" over a path:

- **Arithmetic mean** = best guess for the NEXT single period (forward-looking).
- **Geometric mean** = the REALIZED compound growth you actually earned (backward-looking).

Six Ways Returns Go Wrong in Practice

The traps that separate a clean analysis from a misleading one

Pitfall	What goes wrong	The fix
Adding simple returns over time	Distorts the multi-period total	Multiply gross returns (or use logs)
Quoting the arithmetic mean as return	Ignores vol drag; overstates growth	Use the geometric mean instead
Ignoring dividends	Understates the true total return	Use total-return (adjusted) prices
Annualizing volatility linearly	Scales too fast; inflates risk	Scale by $\sqrt{\text{time}}$, not time
Mixing currencies	Confounds asset return with FX	Convert consistently; state the base
Survivorship in the data	Dead names dropped; looks too good	Use survivorship-free, point-in-time data

From Prices to Returns

No manual coding — you direct Claude Code and read what it writes

GOAL Build a reusable returns pipeline by prompting Claude Code, then read and verify the code it produces.

[INPUT] Prompt · Claude Code

In this repo, create `returns.py` that:

1. downloads daily ADJUSTED prices for a ticker I pass (default AAPL) via `yfinance`, from 2015-01-01 to today;
2. computes simple and log returns;
3. verifies $r_{\log} = \ln(1 + R_{\text{simple}})$ per row;
4. checks that summed monthly log returns equal $\ln(\text{product of gross returns})$;
5. saves `returns.csv` and plots the price, both return series, and their histograms.

Then run it on AAPL and show me the head of `returns.csv` plus the saved figures.

How to work with it

- Read the code — you must understand every line.
- Change the ticker and dates, then re-run.
- If a check fails, tell Claude what you expected.
- Keep `returns.csv` — Labs 2 and 3 reuse it.

Deliverable: `returns.py` + `returns.csv` + the figures, reused across the whole week.

Three Questions Before You Move On

If you can answer these, Part 1 has landed

Q1

A stock returns +20% then -20% (simple). What is the two-day total?

-4%, not 0%

Q2

You want to aggregate one asset's returns over 12 months. Simple or log?

Log (additive)

Q3

Which mean should a fund report for its realized 10-year track record?

Geometric

Cover the right column first, answer, then check. All three follow directly from this part.

Three Things to Carry Forward

Part 1 in one screen

1

Model returns, and default to log returns

Prices are non-stationary; returns are comparable and tractable. Log returns add through time and behave better statistically.

2

Match the mean to the question

Arithmetic mean for the expected next period; geometric mean for realized growth. They differ by the volatility drag, $\approx \sigma^2/2$.

3

Respect the traps

Don't add simple returns over time, don't annualize vol linearly, and always use total-return, survivorship-free data.

Next — Part 2: the Stylized Facts of Returns, the empirical patterns every model must explain.